

The Geographic Gradient of Cardiometabolic Organ Failure

Heart Failure Mortality in the United States 1999–2020

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Introduction and Background

- Cardiometabolic diseases (diabetes, hypertension, obesity, CKD) are major drivers of premature mortality in the United States.
- Epidemiologic research often focuses on upstream risk factors, while less attention is given to the downstream consequences of chronic organ failure.
- Heart failure represents a final common pathway of prolonged metabolic, vascular, and hemodynamic stress.
- Studying heart failure mortality provides a population-level marker of cumulative cardiometabolic injury.
- Understanding geographic and demographic disparities can help identify structural and environmental drivers of disease burden.

Research Question/Objective

- Characterize temporal trends in heart failure mortality in the United States (1999–2020).
- Examine demographic disparities by sex and race/ethnicity.
- Identify geographic variation in heart failure mortality across U.S. states.
- Evaluate the urban–rural gradient in cardiometabolic organ failure mortality.

Methods

Study Design

- Serial cross-sectional analysis of national mortality data.

Data Source

- CDC WONDER Multiple Cause of Death database (1999–2020).

Population

- Adults aged ≥35 years in the United States.

Case Definition

- Heart failure mortality identified using ICD-10 code I50.

Contributing Cardiometabolic Conditions

- Chronic kidney disease (N18)
- Diabetes mellitus (E10–E14)
- Hypertension (I10)
- Obesity (E66)

Analysis

- Age-adjusted mortality rates (AAMRs) per 100,000 population.
- Standardized to the 2000 U.S. population.
- Stratified by year, sex, race/ethnicity, state, and urbanization level.

Results

► Temporal Trends

Heart failure mortality in the United States showed three distinct phases between 1999 and 2020.

- Mortality increased during the early 2000s, followed by a modest decline around 2006–2010.
- After 2012, mortality rose steadily through 2020.
- Mortality rates remained consistently higher in males than in females across all years.

In 2020:

- Male mortality: 5.92 per 100,000
- Female mortality: 4.39 per 100,000

These findings suggest worsening cardiometabolic disease burden in recent years, particularly among men.

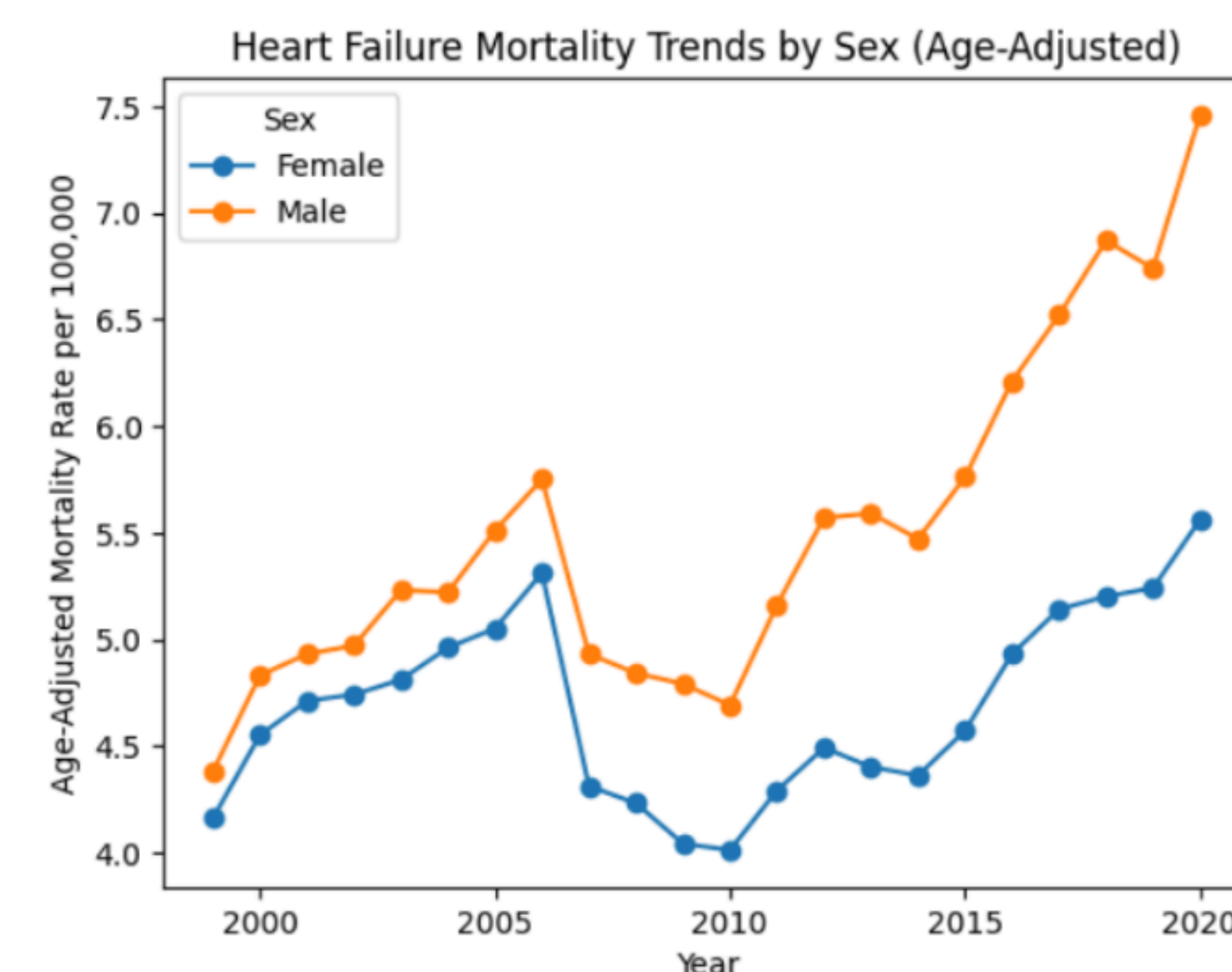


Figure 1. Heart failure mortality trends by sex in the United States, 1999–2020 (age-adjusted). Age-adjusted mortality rates per 100,000 population show higher mortality among males compared with females across all years, with a notable increase after 2012.

► Racial Disparities

Substantial racial and ethnic disparities were observed in heart failure mortality.

- **Black or African American individuals:** highest mortality (6.11 per 100,000)
- **American Indian / Alaska Native:** 3.93 per 100,000
- **White individuals:** 3.88 per 100,000
- **Asian / Pacific Islander:** lowest mortality (2.12 per 100,000)

These disparities persisted across the study period and highlight inequities in cardiometabolic disease progression and outcomes.

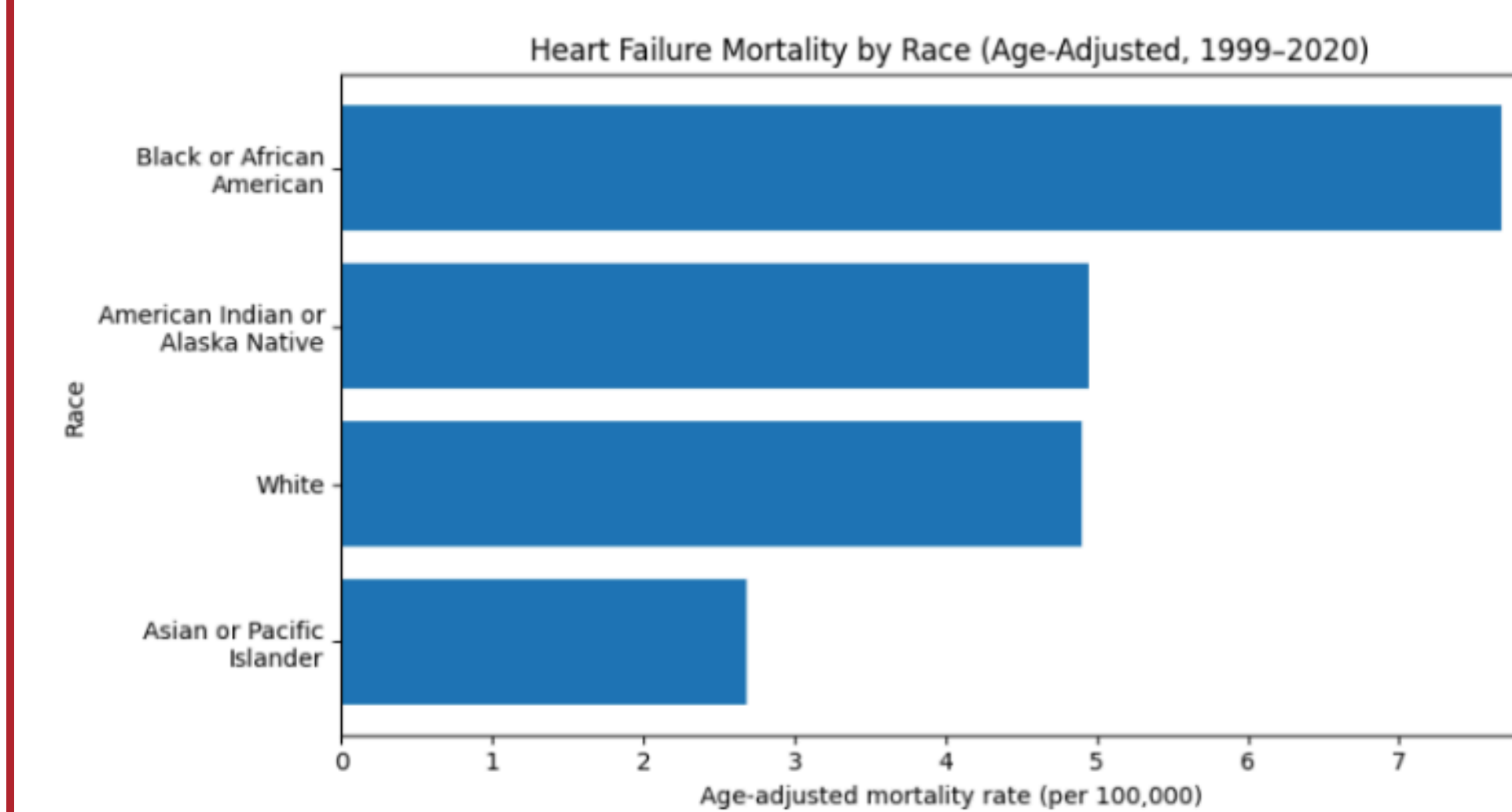


Figure 2. Heart failure mortality by race/ethnicity in the United States, 1999–2020 (age-adjusted). Black individuals experienced substantially higher mortality compared with other racial and ethnic groups, indicating persistent disparities in cardiometabolic disease outcomes.

► Urban–Rural Gradient

Heart failure mortality increased with decreasing urbanization, indicating a strong rural disadvantage.

Age-adjusted mortality rates per 100,000 population:

- **Large metropolitan counties:** 3.52
- **Small metropolitan counties:** 4.51
- **Micropolitan counties:** 5.06
- **Noncore (rural) counties:** 5.52

These findings demonstrate a clear urban–rural gradient in cardiometabolic organ failure mortality, suggesting that structural and healthcare access differences contribute to disparities.

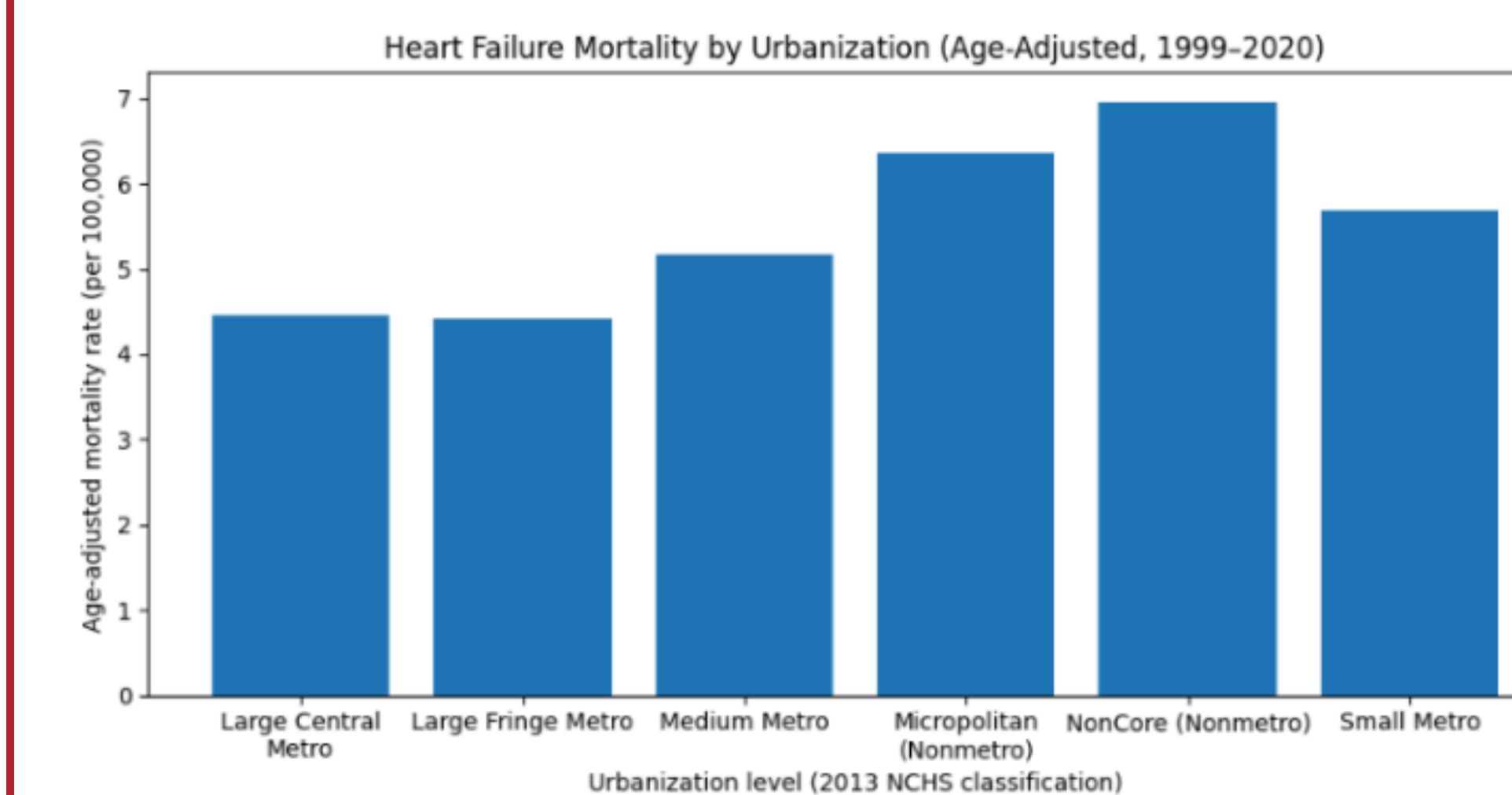


Figure 3. Heart failure mortality by urbanization level in the United States, 1999–2020 (age-adjusted). Mortality rates increased progressively from large metropolitan areas to rural counties, indicating a significant urban–rural gradient in cardiometabolic organ failure mortality.

- Age-adjusted mortality rates (AAMRs) are reported per 100,000 population.
- Rates were standardized to the 2000 U.S. standard population.
- Analyses were stratified by year, sex, race/ethnicity, state, and urbanization level.

Results Continued

► Geographic Variation

Substantial geographic variation in heart failure mortality was observed across U.S. states.

States with the highest mortality rates included:

- Mississippi, Oklahoma, West Virginia

States with the lowest mortality rates included:

- Massachusetts, Arizona, Utah

Overall, mortality rates demonstrated more than twofold variation across states, with the highest rates concentrated in the Southeastern and Appalachian regions.

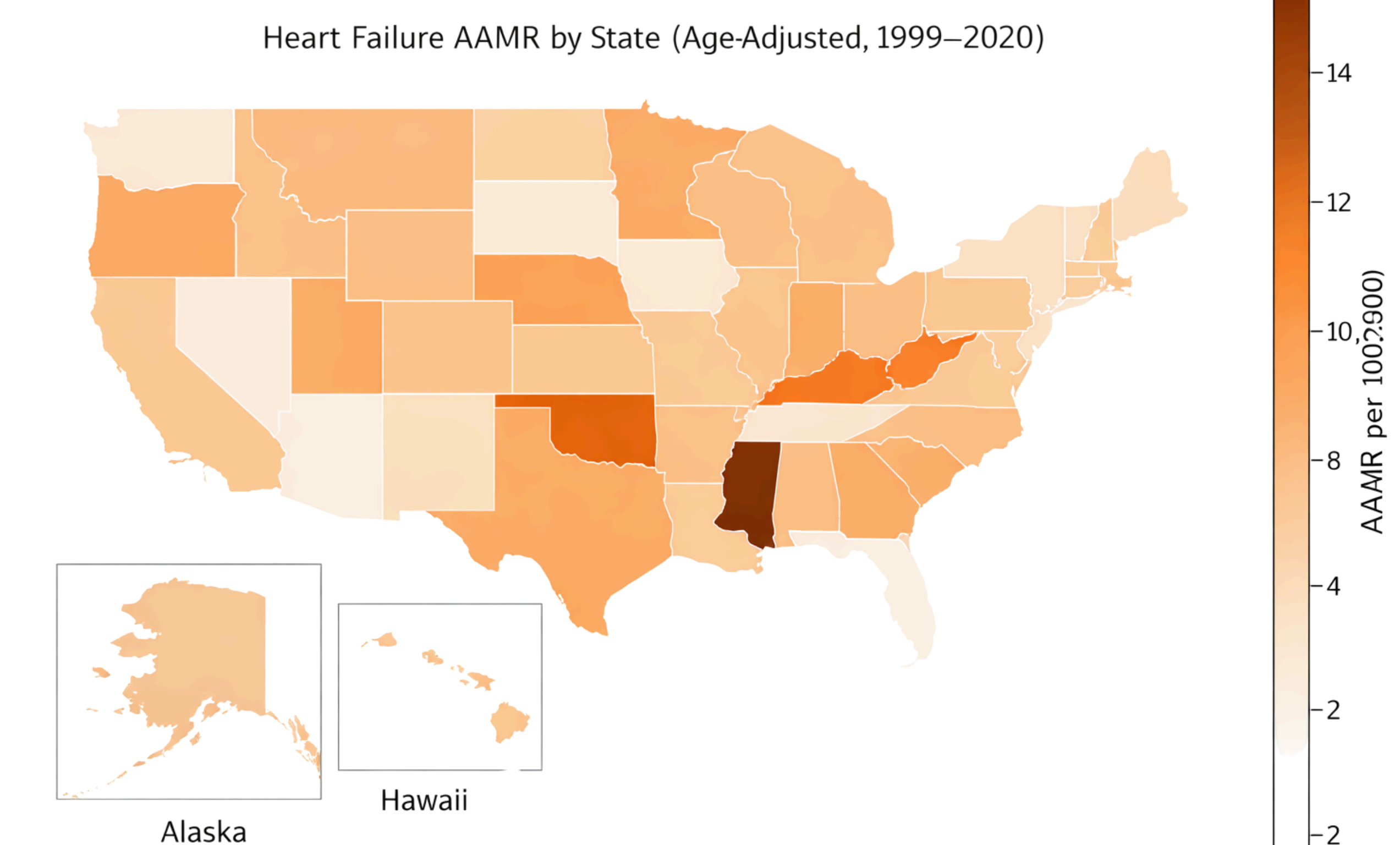


Figure 4. Geographic variation in heart failure mortality across U.S. states (age-adjusted), 1999–2020. State-level mortality rates demonstrate substantial geographic heterogeneity, with the highest burden concentrated in the Southeastern United States.

Conclusion

Heart failure mortality reflects the downstream burden of chronic cardiometabolic disease and reveals marked disparities across the United States. From 1999 to 2020, mortality increased in recent years, remained higher in men, was greatest among Black individuals, and rose progressively across more rural areas, with the highest burden concentrated in the Southeast and Appalachia.

These findings highlight the need for place-based prevention strategies and stronger efforts to address structural and social drivers of cardiometabolic risk.

References

Centers for Disease Control and Prevention. (2024). CDC WONDER. Cdc.gov; CDC.gov. <https://wonder.cdc.gov/>